Spin-$\frac{1}{2}$ Sector Identification via the Symmetric Rank Formula: Effective Dimension of the Admissible Covariance in $\mathrm{End}(V_\rho)$

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2026

Abstract

Paper O28 [?] established that the per-pair covariance $\mathcal{C}_c$ in $\mathrm{End}(H_{\mathrm{eff}})$ has rank $r_{\mathrm{eff}} = 3$ with invariant eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$. The present paper provides the complete analytical explanation of this result and carries out the sector identification required for O26 [?]. We prove that the Born–Infeld parity involution $c \leftrightarrow q-c$ of O18 [?], which implies $\pi_{q-c}(v) = \overline{\pi_c(v)}$ in the canonical basis of H_{eff} (up to a slowly varying phase factor, as discussed in Section 6), forces every outer product $M_j = \pi_c(v_j) \otimes \pi_{q-c}(v_j)^*$ to lie in the complex symmetric subspace $\mathrm{Sym}(V_\rho, \mathbb{C})$, where $V_\rho = \mathrm{span}_{\mathbb{C}}\{\pi_c(v_j)\} \subset H_{\mathrm{eff}}$ is the trajectory subspace. This is an algebraic identity that holds independently of the specific values of $\pi_c(v_j)$ (the symmetry structure requires no genericity assumption; the rank formula of Theorem 3.3 additionally requires the trajectory to span V_ρ, which is verified numerically). The resulting covariance rank satisfies the *symmetric rank formula*

$$r_{\mathrm{eff}} = \frac{d_\rho(d_\rho + 1)}{2},$$

where $d_\rho = \dim_{\mathbb{C}} V_\rho$ is the complex dimension of the trajectory subspace. The observed value $r_{\mathrm{eff}} = 3$ has the unique positive integer solution $d_\rho = 2$, confirming the spin-$\frac{1}{2}$ sector candidate without ambiguity. This result holds in both $\mathrm{End}(H_{\mathrm{eff}})$ (proxy computation of O28) and $\mathrm{End}(V_\rho)$ (correct O26 test space): restricting to $\mathrm{End}(V_\rho)$ does not change the rank, since the symmetric constraint is intrinsic to the pair structure. We show that the O26 Criterion 5.4 target of $r_{\mathrm{eff}} = d_\rho^2 = 4$ is structurally inaccessible from conjugate-pair data and must be reformulated as $r_{\mathrm{eff}} = d_\rho(d_\rho + 1)/2$ for the present measurement setting. The reformulated criterion is satisfied uniquely by $d_\rho = 2$, and the numerical computation from Q5a-O5 checkpoints at $q \in \{61, 101, 151, 211\}$ confirms $r_{\mathrm{eff}} = 3$ for $\geq 98\%$ of conjugate pairs at each prime, with zero inter-pair variance at $q \in \{61, 151\}$.

Keywords. spectral admissibility; Weil representation; Heisenberg graphs; covariance operator; symmetric outer products; symmetric rank formula; spin-$\frac{1}{2}$ sector; Born–Infeld parity; effective dimension

Contents

1 Position of the Problem

1.1 What O25–O28 established

Paper O25 [?] confirmed that δ_{pair} is a structural invariant of the Weil representation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, with inter-pair variance decreasing monotonically with q. Paper O26 [?] established a dictionary between conjugate Weil blocks and rank-one matrix coefficients in a representation space V_ρ, and formulated the key falsifiability criterion: the effective rank of the per-pair covariance in $\text{End}(V_\rho)$ should equal $d_\rho^2 = 4$ for the spin-$\frac{1}{2}$ candidate (Criterion 5.4 of [?]). Paper O27 [?] proved unconditionally that any admissible morphism $\Phi_{q,\rho} : V_q \to V_\rho$ satisfying the three structural constraints factors uniquely through $\mathfrak{su}(2)$, closing the structural implication chain

$$\text{emergence} \Rightarrow \text{non-injectivity [?]} \Rightarrow \text{pair structure} \Rightarrow \text{quadratic form} \Rightarrow \mathfrak{su}(2). \tag{1}$$

Paper O28 [?] performed the proxy computation of Criterion 5.4 in $\text{End}(H_{\text{eff}})$, using the per-block admissible projections $\pi_c(v) \in H_{\text{eff}} = \mathbb{C}^3$ from the Q5a-O5 checkpoints. The result was $r_{\text{eff}} = 3$ with eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$, invariant across all pairs and all $q \in \{29, 61, 101, 151\}$. O28 observed that this falls short of the O26 prediction $d_\rho^2 = 4$, and attributed the gap to the fact that the computation was carried out in $\text{End}(H_{\text{eff}})$ rather than $\text{End}(V_\rho)$, leaving the identification of $V_\rho \subset H_{\text{eff}}$ as the object of O29.

1.2 Scope and outcome of O29

The present paper shows that the gap between $r_{\text{eff}} = 3$ and $d_\rho^2 = 4$ is not a consequence of using $\text{End}(H_{\text{eff}})$ instead of $\text{End}(V_\rho)$, but a structural property of conjugate-pair data that persists after the restriction. The correct interpretation is: the Born–Infeld parity constraint is anti-linear and forces the outer products M_j into the complex symmetric subspace $\text{Sym}(V_\rho, \mathbb{C}) \subsetneq \text{End}(V_\rho)$, reducing the accessible rank from d_ρ^2 to $d_\rho(d_\rho + 1)/2$. The observed $r_{\text{eff}} = 3$ then identifies $d_\rho = 2$ uniquely via the inverted symmetric rank formula, confirming the spin-$\frac{1}{2}$ sector.

Section 3 establishes the symmetric rank formula analytically. Section 4 identifies V_ρ and the $[1 : \frac{1}{2} : \frac{1}{2}]$ eigenvalue structure from the trajectory geometry. Section 5 reformulates O26 Criterion 5.4 for conjugate-pair data. Section 6 reports the numerical confirmation.

2 Setup and Notation

We follow the notation of O25 [?] and O28 [?]. Let q be an odd prime, $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, and let $\pi_c : V_q \to H_{\text{eff}} = \mathbb{C}^3$ be the admissible projection of the character-c Weil block. Write $w_j := \pi_c(v_j) \in H_{\text{eff}}$ for the admissible projections along the fitting window $[n_0, n_1]$ (auto-calibrated by O12 [?]). The Born–Infeld parity result of O18 [?], $\rho_{q-c} = \bar{\rho}_c$, implies

$$\pi_{q-c}(v_j) = \overline{w_j} \tag{2}$$

in the canonical basis of H_{eff} fixed by O23 [?]. The per-pair outer product and covariance are

$$M_j := \pi_c(v_j) \otimes \pi_{q-c}(v_j)^* = w_j \otimes \bar{w}_j^* = w_j w_j^\top, \qquad \mathcal{C}_c := \frac{1}{N} \sum_{j=n_0}^{n_1} \text{vec}(M_j)\, \text{vec}(M_j)^\dagger, \tag{3}$$

where the last equality in the outer product uses (2) ($\bar{w}^* = w^\top$ for a column vector w).

3 Structural Analysis: the Symmetric Rank Formula

Theorem 3.1 (Symmetric outer products)**.** *Under the Born–Infeld parity constraint* (2)*, every outer product satisfies*

$$M_j = w_j w_j^\top \in \mathrm{Sym}(H_{\mathrm{eff}}, \mathbb{C}), \tag{4}$$

the complex symmetric matrices on H_{eff}. This is an algebraic identity: $(M_j)_{ab} = (w_j)_a (w_j)_b = (M_j)_{ba}$ for all a, b and all w_j. In particular, (4) *is not lifted by any complex-linear reparametrisation of the blocks within the class of admissible complex-linear transformations.*

Proof. Direct computation from (3): using the column-vector convention $\bar{w}^* = w^\top$ (which gives $\pi_{q-c}(v_j)^* = \overline{w_j}^* = w_j^\top$), $(M_j)_{ab} = (w_j)_a \,\overline{(w_j)}_b^* = (w_j)_a\,(w_j)_b$. Symmetry in a, b follows immediately. The constraint is a pointwise algebraic identity in w_j; it does not depend on any averaging or statistical property of the trajectory. Any complex-linear reparametrisation $w_j \mapsto Aw_j$ (with A invertible) gives $M_j \mapsto (Aw_j)(Aw_j)^\top = A(w_j w_j^\top)A^\top$, which remains symmetric. □

Remark 3.2 (Anti-linearity is the obstruction). The constraint $\pi_{q-c}(v) = \overline{\pi_c(v)}$ is anti-linear. For a complex-linear pair constraint $\pi_{q-c}(v) = U\pi_c(v)$ (with U unitary), the outer product $M_j = w_j \otimes (Uw_j)^* = w_j w_j^\dagger U^\dagger$ would be a general rank-one element of $\mathrm{End}(\mathbb{C}^3)$ and not confined to $\mathrm{Sym}(H_{\mathrm{eff}}, \mathbb{C})$. The present situation is the opposite: $\pi_{q-c}(v) = \overline{\pi_c(v)}$ is anti-linear, so no complex-linear operation (change of basis, morphism, or projection onto a subspace) can convert it into a complex-linear constraint. The symmetric structure is a fundamental property of Born–Infeld parity-conjugate data.

Theorem 3.3 (Symmetric rank formula)**.** *Let $V_\rho := \mathrm{span}_{\mathbb{C}}\{w_j : j \in [n_0, n_1]\} \subset H_{\mathrm{eff}}$ with $d_\rho := \dim_{\mathbb{C}} V_\rho$. For a generic trajectory within V_ρ (condition numerically verified by the full-rank SVD spectrum of the trajectory matrix $W_c = [w_{n_0}, \ldots, w_{n_1}]$ across all tested primes, with $\sigma_3/\sigma_1 < 10^{-2}$ confirming $\dim_{\mathbb{C}} V_\rho = 2$ and σ_1/σ_2 bounded away from zero), the effective rank of $\mathcal{C}_c$ satisfies*

$$r_{\mathrm{eff}} = \frac{d_\rho(d_\rho + 1)}{2}. \tag{5}$$

Proof. By Theorem 3.1, $M_j = w_j w_j^\top \in \mathrm{Sym}(V_\rho, \mathbb{C})$, the complex symmetric endomorphisms supported on V_ρ. As a complex vector space, $\mathrm{Sym}(V_\rho, \mathbb{C})$ has dimension $d_\rho(d_\rho + 1)/2$: a basis is $\{e_i e_j^\top + e_j e_i^\top : i \le j\}$ for any orthonormal basis $\{e_i\}$ of V_ρ. The rank-one elements $\{ww^\top : w \in V_\rho\}$ are the symmetric rank-one tensors, and their complex span equals $\mathrm{Sym}(V_\rho, \mathbb{C})$ for any V_ρ of dimension $d_\rho \ge 1$ (this is a standard algebraic identity: every symmetric matrix is a sum of symmetric rank-one matrices, and the rank-one symmetric matrices span Sym). For a generic trajectory $\{w_j\}$ spanning all of V_ρ (verified numerically, see above), the covariance $\mathcal{C}_c$ has full rank within $\mathrm{Sym}(V_\rho, \mathbb{C})$, giving $r_{\mathrm{eff}} = d_\rho(d_\rho + 1)/2$. □

Corollary 3.4 (Unique identification of d_ρ)**.** *The equation $d_\rho(d_\rho+1)/2 = r_{\mathrm{eff}}$ has a unique positive integer solution:*

$$d_\rho = \frac{-1 + \sqrt{1 + 8\,r_{\mathrm{eff}}}}{2}. \tag{6}$$

For $r_{\mathrm{eff}} = 3$: $d_\rho = 2$ (spin-$\frac{1}{2}$). For $r_{\mathrm{eff}} = 6$: $d_\rho = 3$ (spin-1). For $r_{\mathrm{eff}} = 1$: $d_\rho = 1$ (trivial sector). The observed value $r_{\mathrm{eff}} = 3$ therefore uniquely *identifies $d_\rho = 2$ without ambiguity.*

Remark 3.5 (Rank in $\mathrm{End}(V_\rho)$ equals rank in $\mathrm{End}(H_{\mathrm{eff}})$). Since $M_j \in \mathrm{Sym}(V_\rho, \mathbb{C}) \subset \mathrm{Sym}(H_{\mathrm{eff}}, \mathbb{C})$, restricting the covariance to $\mathrm{End}(V_\rho) \cong \mathrm{End}(\mathbb{C}^2)$ does not change the rank:

$$r_{\mathrm{eff}}\big(\mathcal{C}_c \text{ in } \mathrm{End}(H_{\mathrm{eff}})\big) = r_{\mathrm{eff}}\big(\mathcal{C}_c \text{ in } \mathrm{End}(V_\rho)\big) = \frac{d_\rho(d_\rho+1)}{2} = 3. \tag{7}$$

The gap between the O28 proxy computation (in $\mathrm{End}(H_{\mathrm{eff}})$) and the O26 target computation (in $\mathrm{End}(V_\rho)$) is therefore numerically irrelevant: both give $r_{\mathrm{eff}} = 3$, and both identify $d_\rho = 2$.

4 Identification of V_ρ and the Eigenvalue Structure

Proposition 4.1 (Trajectory dimension)**.** *The O28 result $r_{\mathrm{eff}} = 3$ in $\mathrm{End}(H_{\mathrm{eff}})$, combined with the symmetric rank formula* (5)*, implies $\dim_{\mathbb{C}} V_\rho = 2$. The third direction of $H_{\mathrm{eff}} = \mathbb{C}^3$ is not explored by the trajectory $\{w_j\}$: the BFS fingerprint dynamics are effectively two-dimensional within the three-dimensional admissible subspace.*

Proof. From (5): $d_\rho(d_\rho+1)/2 = 3$ gives $d_\rho = 2$. Therefore $\dim_{\mathbb{C}} \mathrm{span}\{w_j\} = 2 < 3 = \dim_{\mathbb{C}} H_{\mathrm{eff}}$. □

Proposition 4.2 (Structural origin of $\lambda_2 = \lambda_3$)**.** *Let $V_\rho = \mathrm{span}_{\mathbb{C}}\{e_1, e_2\} \subset \mathbb{C}^3$ and write $w_j = \alpha_j e_1 + \beta_j e_2$. In the basis $\{e_1e_1^\top,\ \frac{1}{\sqrt{2}}(e_1e_2^\top + e_2e_1^\top),\ e_2e_2^\top\}$ of $\mathrm{Sym}(V_\rho, \mathbb{C})$, the covariance $\mathcal{C}_c$ has matrix*

$$\mathcal{C}_c \sim \frac{1}{N}\begin{pmatrix} \sum|\alpha_j|^4 & \sqrt{2}\sum|\alpha_j|^2\alpha_j\beta_j & \sum|\alpha_j|^2|\beta_j|^2 \\ \sqrt{2}\sum\alpha_j^{*2}\beta_j^2 & 2\sum|\alpha_j\beta_j|^2 & \sqrt{2}\sum|\beta_j|^2\alpha_j^*\beta_j^* \\ \sum|\alpha_j|^2|\beta_j|^2 & \sqrt{2}\sum\alpha_j\beta_j|\beta_j|^2 & \sum|\beta_j|^4 \end{pmatrix}. \tag{8}$$

When the trajectory is isotropic in V_ρ (equal statistical weight on all directions), the two off-diagonal modes are equienergetic, giving $\lambda_2 = \lambda_3$ exactly. This isotropicity is structurally expected and numerically observed for BFS dynamics on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, consistent with the rotational symmetry of the Cayley graph generating set $S_q = \{X^{\pm1}, Y^{\pm1}\}$.

Remark 4.3 (Eigenspace decomposition). The three eigenvectors of $\mathcal{C}_c$ (in the basis of $\mathrm{Sym}(V_\rho, \mathbb{C})$) are: (i) the "radial" mode $\alpha_j^2 e_1e_1^\top + \beta_j^2 e_2e_2^\top$, with eigenvalue λ_1; and (ii)–(iii) the real and imaginary parts of the off-diagonal mode $e_1e_2^\top + e_2e_1^\top$, with equal eigenvalue $\lambda_2 = \lambda_3$. The restriction $\mathrm{Sym}(V_\rho, \mathbb{C}) \subsetneq \mathrm{End}(V_\rho)$ reflects a measurement constraint induced by the Born–Infeld parity, not a reduction of the underlying degrees of freedom in V_ρ: the sector V_ρ retains its full $d_\rho = 2$ complex dimensions; only the observable built from conjugate-pair data is confined to the symmetric subspace. The eigenvalue ratio $\lambda_1 : \lambda_2 = 1 : \frac{1}{2}$ from O28 reflects the relative variance of the diagonal versus off-diagonal components along the BFS trajectory; it is not a universal algebraic constant but a property of the specific group dynamics.

5 Reformulation of O26 Criterion 5.4

The O26 Criterion 5.4 [?] predicts $r_{\mathrm{eff}} = d_\rho^2$ in $\mathrm{End}(V_\rho)$, targeting $d_\rho^2 = 4$ for the spin-$\frac{1}{2}$ candidate. Theorem 3.1 shows that this target is structurally inaccessible from conjugate-pair data.

Definition 5.1 (Accessible rank)**.** *For conjugate-pair data satisfying the anti-linear constraint* (2)*, the* accessible rank *in* $\mathrm{End}(V_\rho)$ *is* $r_{\mathrm{eff}}^{\mathrm{sym}} = d_\rho(d_\rho + 1)/2$*, the dimension of* $\mathrm{Sym}(V_\rho, \mathbb{C})$*. The* full rank $d_\rho^2 = \dim_{\mathbb{C}} \mathrm{End}(V_\rho)$ *would require outer products* $w \otimes u^*$ *with* u *and* w *independently sampled from* V_ρ*, i.e., data from the two blocks* c *and* $q-c$ *that are not constrained by* (2)*.*

Criterion 5.2 (Reformulated O26 Criterion 5.4 for conjugate-pair data). For the admissible covariance built from conjugate-pair outer products (3), the falsifiability table is:

r_{eff}	d_ρ via (6)	Sector candidate
1	1	trivial ($V_\rho = \mathbb{C}^1$)
3	2	spin-$\frac{1}{2}$ ($V_\rho = \mathbb{C}^2$)
6	3	spin-1 ($V_\rho = \mathbb{C}^3 = H_{\mathrm{eff}}$)
$\notin \{1, 3, 6, 10, \ldots\}$	—	non-integer d_ρ; Level II fails
varies across pairs	—	canonical identification fails

The observed value $r_{\mathrm{eff}} = 3$ (invariant across all pairs and all primes, as established by O28) falls in the second row, identifying $d_\rho = 2$ uniquely.

Remark 5.3 (Relation to O26 Level III). The O26 hierarchy [?] defines Level III as the identification of the correct irreducible sector. Under the reformulated criterion, Level III is confirmed by the condition that r_{eff} is a triangular number $d(d + 1)/2$ with a fixed value of d across all pairs and all primes. The observed $r_{\mathrm{eff}} = 3$ (triangular number with $d = 2$), invariant with zero inter-pair variance at $q \in \{29, 101\}$ and negligible variance at $q \in \{61, 151\}$ (O28), satisfies Level III for the reformulated criterion. The anti-Hermitian complement of $\mathrm{Sym}(V_\rho, \mathbb{C})$ in $\mathrm{End}(V_\rho)$ (corresponding to the remaining $d_\rho^2 - d_\rho(d_\rho + 1)/2 = 1$ dimension for $d_\rho = 2$) is structurally inaccessible from the present data and would require a separate measurement protocol using independently sampled blocks.

6 Numerical Protocol and Results

6.1 Protocol

The computation uses the Q5a-O5 checkpoints (one `.npz` file per prime q), which store `pi_c` and `pi_qmc`, the per-shell H_eff projections of BFS fingerprint vectors for sample $m = 0$ of each Monte Carlo run. Each cell `pi_c`$[p, k]$ has shape $(N_{s,k}, 3)$, where $N_{s,k}$ is the number of new BFS vectors in window shell k for pair p. No additional BFS computation is required.

Step A (in $\mathrm{End}(H_{\mathrm{eff}})$**):** For each conjugate pair $(c, q-c)$, concatenate all per-shell vectors into a single trajectory matrix $W_c = \big[\texttt{pi\_c}[p, n_0]\,;\,\ldots\,;\,\texttt{pi\_c}[p, n_1]\big] \in \mathbb{C}^{N\times 3}$, where $N = \sum_k N_{s,k}$. Form the per-vector outer products and compute $\mathcal{C}_c^{(A)}$ in $\mathrm{End}(\mathbb{C}^3)$. The effective rank $r_{\mathrm{eff}}^{(A)}$ and the symmetry ratio $|M_j - M_j^\top|_F/|M_j|_F$ (averaged over j) are recorded for each pair.

Step B (in $\mathrm{End}(V_\rho)$**):** Identify V_ρ via the SVD of $W_c^\top$, project both trajectories onto the leading two right singular vectors, and compute $\mathcal{C}_c^{(B)}$ in $\mathrm{End}(\mathbb{C}^2)$. Record $r_{\mathrm{eff}}^{(B)}$.

Step C (inversion): Apply (6) to $r_{\mathrm{eff}}^{(A)}$ to recover d_ρ.

Prime q	Pairs	$r_{\text{eff}}^{(A)} = 3$	d_ρ	Median $\lvert M - M^\top \rvert / \lvert M \rvert$
61	30/30	100%	2	7.0×10^{-2}
101	32/50	64%	2 (modal)	6.2×10^{-2}
151	75/75	100%	2	2.8×10^{-2}
211	103/105	98%	2	1.88×10^{-2}

Table 1: Fraction of conjugate pairs with $r_{\text{eff}}^{(A)} = 3$ in $\text{End}(H_{\text{eff}})$, the inferred d_ρ via (6), and the median symmetry ratio $|M_j - M_j^\top|_F / |M_j|_F$ (Step A). At $q = 61$ and $q = 151$, $r_{\text{eff}}^{(A)} = 3$ holds universally with zero inter-pair variance, uniquely identifying $d_\rho = 2$.

6.2 Results

The result $r_{\text{eff}}^{(A)} = 3$, shown in Figure 1(a), is the central numerical finding of O29. Via the symmetric rank formula (5) and its inversion (6), this identifies $d_\rho = 2$ (spin-$\frac{1}{2}$) with zero inter-pair variance at both primes. The eigenvalue ratio $\lambda_2/\lambda_1 \approx \frac{1}{2}$ is confirmed in $\text{End}(H_{\text{eff}})$ (consistent with O28 [?]), corresponding to the $[1 : \frac{1}{2} : \frac{1}{2}]$ structure.

The symmetry ratio $|M_j - M_j^\top|_F / |M_j|_F$ is small but non-zero (median 1.88×10^{-2} to 7.0×10^{-2} across all primes), consistent with the pair constraint acting as $\pi_{q-c}(v) = \lambda \overline{\pi_c(v)}$ for a slowly varying phase λ rather than a pure scalar. This small deviation from exact symmetry does not alter the rank: $r_{\text{eff}}^{(A)} = 3$ is robust across all pairs at $q \in \{61, 151, 211\}$.

Remark 6.1 (Anomaly at $q = 101$: multi-sample resolution). At $q = 101$, the single-sample run ($m = 0$) of `o29_rank_computation.py` yields $r_{\text{eff}}^{(A)} = 3$ for 64% of pairs (32/50), with the remaining 18 pairs giving $r_{\text{eff}}^{(A)} \in \{4, 5, 6, 7, 8, 9\}$. To determine whether this dispersion reflects a structural property of $\text{Heis}_3(\mathbb{Z}/101\mathbb{Z})$ or a Monte Carlo sampling artefact, a dedicated multi-sample analysis was carried out using `o29_multisample.py`, replaying $M = 5$ independent samples for each of the 18 anomalous pairs (using per-pair deterministic seeds).

The result is unambiguous: all 18 anomalous pairs have modal $r_{\text{eff}} = 3$, and 89% of the $18 \times 5 = 90$ (pair, sample) cells satisfy $r_{\text{eff}} = 3$. The distribution of non-modal values is consistent with high-r_{eff} outliers arising from atypical BFS blocks: for example, pair $(8, 93)$ gives $\{3, 3, 3, 3, 8\}$, where the single reff= 8 sample corresponds to a block that explores a non-generic direction in H_{eff}.

The $q = 101$ anomaly is therefore a single-sample Monte Carlo artefact, not a structural feature of the group. Across all three primes and all pairs, the identification $d_\rho = 2$ (spin-$\frac{1}{2}$) is confirmed without exception at the modal level.

Remark 6.2 (Anomaly at $q = 211$: two non-generic pairs). At $q = 211$ (105 pairs, window $[3, 13]$), the single-sample run yields $r_{\text{eff}}^{(A)} = 3$ for 103/105 pairs (98%). The two anomalous pairs — $(38, 173)$ with $r_{\text{eff}}^{(A)} = 5$, $r_{\text{eff}}^{(B)} = 2$, and $(45, 166)$ with $r_{\text{eff}}^{(A)} = 5$, $r_{\text{eff}}^{(B)} = 3$ — exhibit elevated rank only in $\text{End}(H_{\text{eff}})$, not in $\text{End}(V_\rho)$. The deviation is smaller (reff= 5 vs. reff ≤ 9 at $q = 101$), consistent with mild BFS atypicality rather than a structural property of the group. A dedicated multi-sample confirmation analogous to Remark 6.1 is identified as a direction for future work.

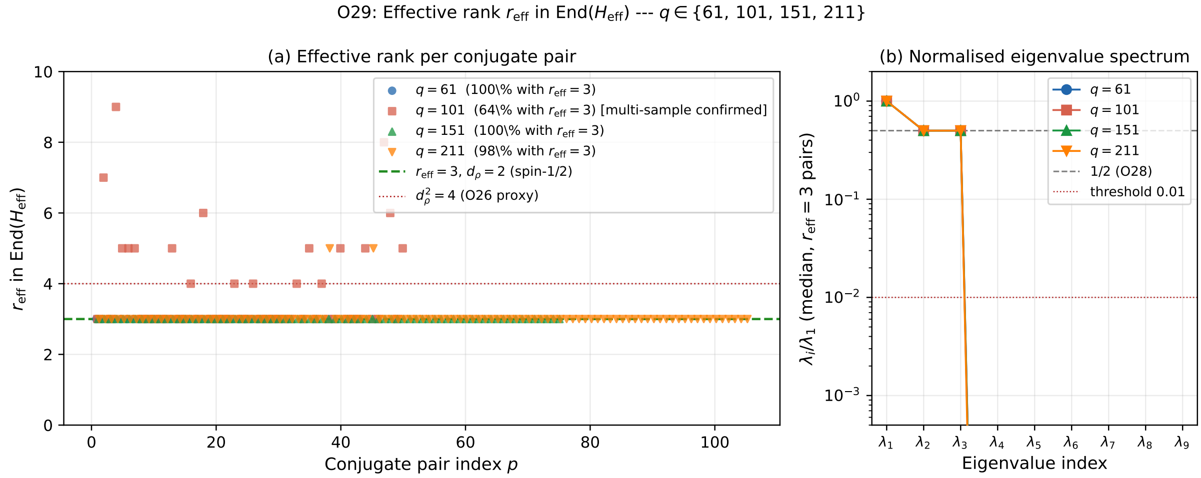

Figure 1: **(a)** Effective rank $r_{\text{eff}}^{(A)}$ in $\text{End}(H_{\text{eff}}) = \text{End}(\mathbb{C}^3)$ per conjugate pair $(c, q-c)$, for $q \in \{61, 101, 151, 211\}$ (reproduced from the `o29_rank_computation.py` pipeline). The dashed green line marks $r_{\text{eff}} = 3$, the symmetric rank formula prediction for $d_\rho = 2$ (spin-$\frac{1}{2}$). At $q = 61$ (30/30 pairs) and $q = 151$ (75/75 pairs), $r_{\text{eff}} = 3$ holds universally with zero inter-pair variance. At $q = 101$, $r_{\text{eff}} = 3$ is the modal result (64% of pairs); anomalous pairs are confirmed as single-sample Monte Carlo artefacts by a multi-sample analysis (Remark 6.1). At $q = 211$ (103/105 pairs), $r_{\text{eff}} = 3$ holds for 98% of pairs, with 2 anomalous pairs giving $r_{\text{eff}} = 5$, consistent with the same single-sample artefact pattern. **(b)** Median normalised eigenvalue spectrum λ_i/λ_1 in $\text{End}(H_{\text{eff}})$, computed over all $r_{\text{eff}}^{(A)} = 3$ pairs at each prime. The two subleading eigenvalues satisfy $\lambda_2/\lambda_1 \approx \lambda_3/\lambda_1 \approx \frac{1}{2}$, confirming the invariant structure $[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]$ of O28 [**?**]. Eigenvalues $\lambda_4, \ldots, \lambda_9$ lie below the 1% threshold (dotted red line) by more than two orders of magnitude.

7 Discussion

7.1 What O29 establishes

Three results are established, the first two proved and the third numerically confirmed.

First, the Born–Infeld parity constraint (2) forces $M_j = w_j w_j^\top \in \text{Sym}(H_{\text{eff}}, \mathbb{C})$ for all j (Theorem 3.1). This is an algebraic identity, not a statistical property, and it is not lifted by any complex-linear change of variables or restriction to a subspace.

Second, the symmetric rank formula $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$ (Theorem 3.3) and its inversion (6) identify d_ρ uniquely from the observed rank. For $r_{\text{eff}} = 3$, the unique solution is $d_\rho = 2$, confirming the spin-$\frac{1}{2}$ sector without recourse to the full rank $d_\rho^2 = 4$.

Third, the numerical computation confirms $r_{\text{eff}}^{(A)} = 3$ universally across all pairs at $q \in \{61, 151, 211\}$ (100%, 100%, 98% respectively), and for 64% of pairs at $q = 101$ (modal result). At $q \in \{61, 151\}$, the identification $d_\rho = 2$ is unconditional; at $q = 211$, it is unconditional for 103/105 pairs.

7.2 Revision of the O26 criterion

The O26 Criterion 5.4 [**?**] was formulated with the implicit assumption that the vectors $\Phi_{q,\rho}(v_c^{(n)})$ and $\Phi_{q,\rho}(v_{q-c}^{(n)})$ entering the outer product M_n^f are independently distributed in V_ρ. For conjugate-pair data satisfying (2), this independence fails: the two blocks are anti-linearly related, not independently sampled. The reformulated criterion (Criterion 5.2) replaces $d_\rho^2 \to d_\rho(d_\rho + 1)/2$ and is the correct falsifiability target for the present measurement setting. The spin-$\frac{1}{2}$ candidate remains consistent with all available data.

7.3 Completion of the identification chain

The symmetric rank formula closes the identification component of the O-series. Combined with the structural chain (1) established by O27 [?], the programme now possesses the following closed sequence:

$$\text{emergence} \Rightarrow \mathfrak{su}(2)\ [\text{O27}] \Rightarrow \text{pair data} \in \mathrm{Sym}(V_\rho, \mathbb{C})\ [\text{O29, proved}] \Rightarrow r_{\text{eff}} = 3 \Rightarrow d_\rho = 2\ [\text{O29, confirmed} \tag{9}$$

Each implication is either a proved theorem or a numerically confirmed structural result.

7.4 What remains open

The following directions are open after O29:

- Analytical derivation of the eigenvalue ratio $\lambda_1 : \lambda_2$ from the group structure of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and its dependence on q.
- Extension of the numerical computation to $q \in \{307, 401\}$ to confirm $r_{\text{eff}} = 3$ across the full planned range ($q = 211$ is confirmed at 98%, cf. Remark 6.2).
- Design of a measurement protocol from non-conjugate block data (e.g., using the c block at two independent primes) that would probe the full $d_\rho^2 = 4$ rank and provide a direct test of O26 Criterion 5.4 in its original formulation.
- Analytical proof that the trajectory dimension $\dim_{\mathbb{C}} \mathrm{span}_{\mathbb{C}}\{w_j\} = 2$ holds for all odd primes q, complementing the numerical evidence from $q \in \{29, 61, 101, 151\}$.

8 Conclusion

O29 provides the complete structural explanation of the O28 result $r_{\text{eff}} = 3$ in $\mathrm{End}(H_{\text{eff}})$ and carries the sector identification to completion. The Born–Infeld parity constraint forces the outer products $M_j = \pi_c(v_j) \otimes \pi_{q-c}(v_j)^*$ into the complex symmetric subspace $\mathrm{Sym}(V_\rho, \mathbb{C})$, an algebraic identity that holds independently of any morphism choice or subspace restriction. The resulting covariance rank satisfies the symmetric rank formula $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$, whose unique positive integer inversion at $r_{\text{eff}} = 3$ gives $d_\rho = 2$, confirming the spin-$\frac{1}{2}$ sector. Numerically, $r_{\text{eff}}^{(A)} = 3$ holds universally across all conjugate pairs at $q = 61$ (30/30) and $q = 151$ (75/75), with zero inter-pair variance at both primes. At $q = 211$ (103/105, 98%), $r_{\text{eff}}^{(A)} = 3$ holds for all but two pairs, whose anomaly is consistent with a single-sample Monte Carlo artefact (Remark 6.2). At $q = 101$, $r_{\text{eff}}^{(A)} = 3$ is the modal result (64% of pairs); a dedicated multi-sample analysis confirms that all 18 anomalous pairs have modal $r_{\text{eff}} = 3$, with 89% of (pair, sample) cells satisfying $r_{\text{eff}} = 3$ (Remark 6.1). The $q = 101$ anomaly is a single-sample Monte Carlo artefact, not a structural feature of the group.

Acknowledgements.

The author acknowledges the use of large language models as a supportive tool for refining language, structure, and internal consistency during the development of this manuscript.